NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

ACADEMIC YEAR 2026-2027

Assignment

Instructor: Dr. Sivalingam S M
Maximum Marks: 5

Date: 13 November 2026
Submission: 16 November 2026

Instructions: Answer all questions. Show the necessary intermediate steps and calculations. Submit handwritten solutions on or before 16 November 2026.

1. Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Find the eigenvalues of A. Hence, using the Cayley–Hamilton theorem, obtain A^{-1}. **[1 Mark]**

2. Using Taylor's theorem, expand

$$f(x) = \log(1 + x)$$

about $x = 1$ up to term containing $(x-1)^3$. Hence, obtain an approximation for $\log(2.1)$. **[1 Mark]**

3. If

$$u = x^2y + y^2z + z^2x,$$

where

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = r^2,$$

find $\dfrac{\partial u}{\partial r}$ using the chain rule and verify your result by direct substitution. **[1 Mark]**

4. Using the method of Lagrange multipliers, find the maximum and minimum values of

$$f(x, y, z) = xy + yz + zx$$

subject to the constraint

$$x^2 + y^2 + z^2 = 3.$$

[1 Mark]

5. Solve the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = xe^{2x} + \sin 2x.$$

Obtain both the complementary function and the particular integral. **[1 Mark]**